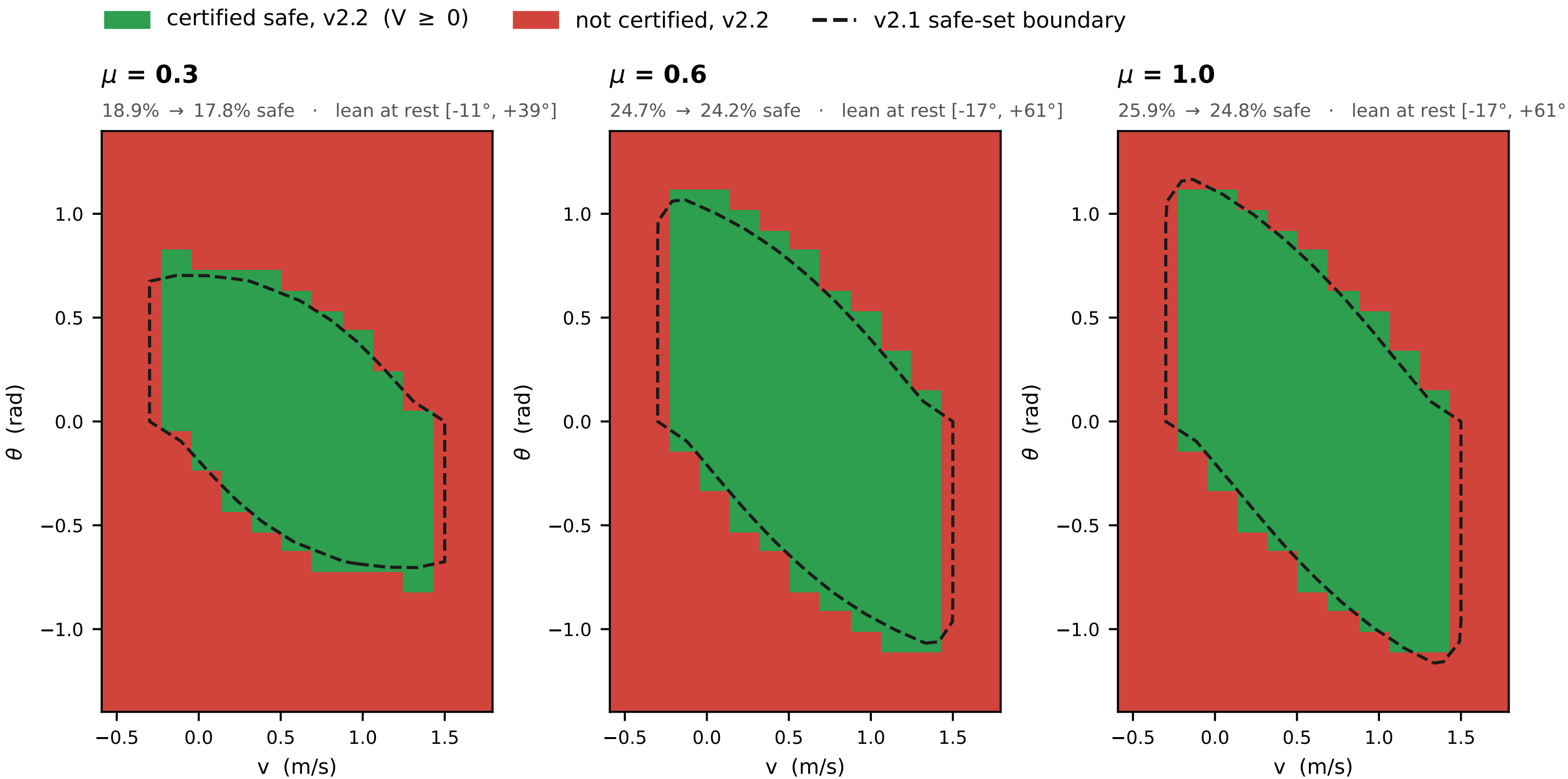
# Certified safe set under the v2.1 -> v2.2 model update

Same solver, same ODD contract, same axes. Only the robot model changed.

The camera-inclusive v2.2 robot is 1.4% heavier with a 4% higher centre of mass. The certified volume shrinks slightly at every friction — $\mu$=0.3: 18.9% → 17.8%, $\mu$=0.6: 24.7% → 24.2%, $\mu$=1.0: 25.9% → 24.8% — but this is not a uniform inward shift. At $\mu \geq 0.6$ the at-rest lean band actually widens (e.g. $\mu$=0.6: [-11°, +55°] → [-17°, +61°]) while the set loses volume elsewhere, so the boundary moves both ways. The overall shape is preserved, which is what a mass and inertia update alone should look like.




Slices at $\dot{\theta} = \dot{\psi} = 0$. Grids: v2.1 read from git at 2c4f73d; v2.2 is the committed regeneration, same contract.
Reproduce: python vault/tools/make_model_update_figure.py